

WasteWise AI — Standard Operating Procedure

Production Waste, Loss, and Rework Management

Document ID: WW-SOP-001

Version: 1.0

Department: Manufacturing / Quality

1. Purpose

This procedure defines how production teams identify, record, investigate, and reduce material loss, nonconformance, and rework during manufacturing.

2. Scope

This SOP applies to all production lines, shifts, products, and manufacturing processes where material usage, waste, loss, or rework is recorded.

3. Waste Classification

Production losses shall be classified as process loss, material loss, nonconformance, rework, recovered material, or final loss. Recovered material must be recorded separately from final loss.

4. Data Recording

Operators shall record the production date, product, production line, shift, process, event reason, production quantity, and loss quantity for each relevant event.

5. Investigation Procedure

When waste exceeds the normal operating level, the supervisor shall review the production line, shift, process, and event reason. Investigation should focus on combinations producing the highest loss quantity or waste rate.

6. Rework Procedure

Nonconforming material that can be corrected shall be classified as rework. Rework quantities must be recorded and linked to the originating production event.

7. Corrective Action

For significant or recurring losses, identify the likely cause, document corrective action, assign an owner, and monitor subsequent production results.

8. Escalation Rules

Escalate an event when loss is unusually high, the same reason occurs repeatedly, a production line shows a sustained increase in waste rate, or a quality issue affects multiple batches.

9. Review Frequency

Production waste data should be reviewed daily by the production team and summarized weekly by the supervisor. Persistent high-loss patterns should be reviewed monthly.

10. Key Performance Indicators

Recommended indicators include total production, total waste, waste rate, loss by production line, loss by product, loss by shift, loss by process, and top waste reasons.

11. Example Investigation Triggers

High waste rate → review process and line conditions. Repeated defect reason → investigate recurring root cause. High rework quantity → review nonconformance. Sudden loss increase → compare recent production history. High loss on one shift → review shift-specific conditions.

End of document. Synthetic test SOP for the WasteWise AI RAG pipeline.